

Appendix of EMA

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A THEORETICAL GUARANTEES OF EMA

In this section, we show that EMA preserves the asymptotic complexity of the base index in terms of both construction time and index size. We also derive the local deletion-work bound stated in Section 4.1 of the main paper. We further analyze the source of false positives introduced by Markers and provide theoretical bounds on the false positive rate.

A.1 Complexity

LEMMA A.1 (COST OF MEncode AND EDGE EXISTANCE). *In Algorithm 3, MEncode takes $O(m)$ time. Testing whether $e_{(n,v)}$ previously exists takes $O(M)$ time, since it scans at most M previously stored neighbors.*

THEOREM A.2. *The time complexity of Algorithm 3 is*

$$O(efc \cdot M \cdot (M + m)).$$

PROOF. The outer loop iterates over candidates $v \in \text{Cand}$, so it executes at most $|\text{Cand}| \leq efc$ times.

For a fixed candidate v , the inner loop scans $u \in \text{Nbrs}$. The neighbor list is bounded by the maximal size M , hence $|\text{Nbrs}| \leq M$ and the inner loop runs at most M times.

Consider one inner-loop iteration for a pair (u, v) . Besides a constant-time distance comparison, when u dominates v the algorithm (i) tests whether $e_{(n,v)}$ previously exists, which costs $O(M)$, and then (ii) calls either Marker merge or MEncode, which costs $O(m)$ by Lemma A.1. Therefore, each inner-loop iteration costs $O(M + m)$ in the worst case.

Thus, for each candidate v , the inner loop costs

$$O(M \cdot (M + m)).$$

The remaining work per candidate (insertion into Nbrs and one additional MEncode) is at most $O(M + m)$ and is dominated by the inner-loop bound. Summing over all $|\text{Cand}| \leq efc$ candidates gives

$$O(efc \cdot M \cdot (M + m)).$$

□

The time complexity of HNSW index construction is empirically close to $O(n \log n)$ and is highly correlated with the parameters M and efc , which can be expressed as $O(M \cdot efc \cdot n \log n)$. In our

method, Marker propagation introduces additional operations, resulting in a complexity of $O(M \cdot (M + m) \cdot efc \cdot n \log n)$. However, since both M and m are constants that are significantly smaller than n , the asymptotic time complexity remains unchanged.

THEOREM A.3. *The space complexity of EMA is*

$$O(nM \cdot sm),$$

PROOF. The additional storage introduced by EMA consists of the Marker stored per neighbor (edge), which is only used in the bottom layer.

In the bottom layer, each of the n nodes stores at most M neighbors, resulting in at most nM edges. Each edge stores a Marker of size sm bits, so the total Marker storage is

$$(nM) \cdot sm \text{ bits.}$$

The top layer contains at most n nodes, each with at most M outgoing edges. It stores no Markers and contributes $O(nM)$ space.

Combining both parts yields a total space complexity of

$$O(nM \cdot sm + nM),$$

which simplifies to $O(Mn \cdot sm)$. □

Deletion complexity. We analyze the local graph and Marker maintenance procedure for deleting one point a in the main paper's Marker-aware Deletion algorithm. Let n denote the number of stored points, including retained deleted records. We retain the degree parameter M , search width efc , and attribute count m . Vector dimension, per-attribute encoding capacities, and the reconnection budgets $k_r = 50$, $c = 3$ are fixed. Thus a distance evaluation costs $O(1)$, while a full Marker operation or MEncode costs $O(m)$. We use the parameterized search model of the construction analysis: navigation and each layer search examine $O(efc \log n)$ rows of degree $O(M)$, at cost $O(M \cdot efc \log n)$ including bounded-queue bookkeeping. Identifier-keyed work lists use the standard expected amortized $O(1)$ model for dictionary lookup and insertion. We count the logical graph and Marker procedure; request compatibility can be evaluated directly over its bounded witness context, without requiring memoization.

LEMMA A.4 (DELETION COMPLEXITY). *Under the stated search and dictionary models, the expected work of one local deletion is*

$$O(M \cdot efc \cdot (M + efc \cdot m) \log n). \quad (1)$$

PROOF. *Search and discovery.* IP-DiskANN's Algorithm 5 searches with the old vector and checks examined adjacency lists for incoming edges [1]. By the adopted model, the constant number of layer searches and incoming-edge scans costs $O(M \cdot efc \log n)$. At most $O(efc \log n)$ incoming sources are discovered. The outgoing pass issues $O(cm)$ proposals, but their source IDs all come from the same first k_r retained candidates, so it adds only $O(1)$ distinct source rows. Proposals are merged by source and layer, and each repair plan executes once. Thus the number r_a of reconstructed adjacency rows across both layers is $O(efc \log n)$. This bounds discovered repair rows, not the deleted point's full in-degree.

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Work per repaired row. Each source receives $O(M)$ old targets, at most c incoming-side replacements, and $O(M)$ outgoing-side targets, giving an $O(M)$ target pool. Selecting replacements scores at most the first k_r retained candidates, rather than launching a fresh graph search for each source. Pool formation, duplicate checks, ranking, and RNG comparisons cost $O(M^2)$ per row. Here pruning records direct witness IDs rather than propagating historical Markers; each pruned candidate is assigned to at most one owner. Coverage admission, old-Marker preservation, and direct-witness encoding therefore add $O(Mm)$. All base-layer repairs share the deletion's witness context, which contains at most efc candidates. For $O(M)$ owners, testing request compatibility and coverage against those witnesses costs $O(M \cdot efc \cdot m)$. The total per repaired row is consequently $O(M^2 + M \cdot efc \cdot m)$.

Context and nonincident Marker cleanup. Encoding the retained candidates and building their support lists costs $O(efc \cdot m)$. Cleanup applies to at most efc retained source rows. Each such source inspects $O(M)$ neighbors to select one dominator, then checks that owner's removable bits against at most efc witnesses. This costs $O((M + efc)m)$ per source and $O(efc(M + efc)m)$ in total.

Aggregation. Identifier lookup and work-list management use the stated expected constant-time dictionary model; target-pool edits have already been charged above. Publishing each row and its Markers costs $O(Mm)$, also covered by the per-row bound. There are no concurrent point workers or deferred inter-point repair searches for this single-point procedure. Writing T_a for the expected local work and substituting $r_a = O(efc \log n)$ gives

$$\begin{aligned} T_a &= O\left(M \cdot efc \log n + efc(M + efc)m\right. \\ &\quad \left.+ (1 + r_a)(M^2 + M \cdot efc \cdot m)\right) \\ &= O\left(M \cdot efc \cdot (M + efc \cdot m) \log n\right), \end{aligned}$$

which proves Equation (1). $\square$

For fixed M , efc , and m , this is $O(\log n)$. As in the construction analysis, this is a local-algorithm bound under the stated models, not an unconditional worst-case traversal or implementation-level guarantee.

Reconnection details. Each incoming- or outgoing-side proposal ranks the original first $k_r = 50$ retained candidate positions and admits at most $c = 3$ eligible endpoints. Invalid or retired endpoints, the deleted point, and the affected endpoint are excluded without widening that prefix. The deleted point's native placeholder still consumes its original rank.

RNG dominance and Marker cleanup. On a Marker-bearing layer, let $F(x) = \text{MEncode}(x.A, C)$ under the fixed Codebook C , and let Cand denote the retained layer-search candidates. For source u , neighbor v , and candidate w , define

$$D(u, v, w) \equiv \text{dis}(u, v) \leq \text{dis}(u, w) \wedge \text{dis}(v, w) < \text{dis}(u, w).$$

The first comparison captures the distance ordering in RNG pruning; the second is its domination test. Nonincident Marker cleanup applies only to retained live candidate sources; examined-only rows may receive graph repair without this cleanup. For a deleted point a and each such source u , select the live neighbor v closest to u

among those satisfying $D(u, v, a)$. If no such neighbor exists, no nonincident Marker is cleaned for that source. Equal-distance ties between valid cleanup owners follow adjacency-row order. The candidate bits for clearing are

$$e_{(u,v)}.\text{Marker} \wedge F(a) \wedge \neg F(v).$$

Clear a set bit b of this mask only when no live $w \in \text{Cand} \setminus \{a, u, v\}$ has both $F(w)[b] = 1$ and $D(u, v, w)$. Thus the target's own encoding and bits supported by retained live witnesses are protected. Reconstruction uses the same $D(u, v, w)$ when testing whether a selected owner already covers $F(w)$ and when choosing a compatible requesting owner for w . This is a current geometric compatibility test, not a recovery of exact historical pruning assignments.

Batched deletion and cross-point repair. The main paper's Marker-aware Deletion algorithm describes one logical point deletion. The batched implementation may execute several such point tasks in parallel. A repair can encounter another target that is being retired by a different task. It hands the source ID, layer, and optional Marker-recovery request to the active deletion task for that target. The recipient applies incoming repair deltas using its retained layer candidates. Closing a task is synchronized with outstanding producers so that reserved requests are not lost. Valid requests that arrive after the recipient has closed enter a batch-local deferred queue; completion epochs distinguish overlapping work from already-completed stale edges.

Deferred repairs may search again at width efc to construct a fresh bounded witness context and can generate further repair handoffs. Each producing worker services deferred work before returning, and the batch waits for its workers and their claimed repairs to finish. These repairs complete within the deletion call rather than waiting for a subsequent query. Liveness is rechecked when using witnesses and publishing edges. Callers must prevent external queries and mutations from overlapping this stop-the-world operation; parallelism is internal to the deletion batch. The single-point complexity bound above does not include this cross-point coordination or its additional repair searches.

A.2 Theoretical Analysis of Marker False Positives

Definition A.5 (Expected Case-(1) False Positive Rate). Consider a directed edge $e_{(u,v)}$ whose Marker is constructed by aggregating the attribute bitmaps of the dominating node v and the set of nodes dominated by v with respect to u , denoted by $D(e)$; hence $v \notin D(e)$.

Let R be a query predicate with selectivity sel , i.e., for a randomly drawn node p ,

$$\Pr[p \models R] = \text{sel},$$

and assume that, conditional on the graph, predicate satisfaction events are independent and identically distributed across nodes.

A *Case-(1) false positive* occurs on edge e if the Marker is activated due to at least one dominated node satisfying the predicate, while the target node itself does not satisfy the predicate. Formally,

$$\text{FP}_{\text{case1}}(e) \iff (v.A \not\models R) \wedge (\exists p \in D(e) : p \models R).$$

This event requires a predicate-matching dominated node; it does not count aggregate matches assembled from different nodes across predicate components.

THEOREM A.6 (EXPECTED CASE-(1) FALSE POSITIVE RATE). Let $|D(e)|$ denote the dominate size of edge e , and let

$$\mu = \mathbb{E}[|D(e)|]$$

be the average dominate size over all edges.

Under the assumptions of Definition A.5, the expected false positive rate induced by Case (1) is given by

$$\text{FPR}_{\text{case1}} = (1 - \text{sel}) \cdot \left(1 - \mathbb{E}[(1 - \text{sel})^{|D(e)|}]\right).$$

Moreover, the false positive rate admits the following upper bound that depends only on the average dominate size μ :

$$0 \leq \text{FPR}_{\text{case1}} \leq (1 - \text{sel}) \cdot (1 - (1 - \text{sel})^\mu).$$

In particular, under the common average-size approximation $|D(e)| \approx \mu$, we obtain

$$\text{FPR}_{\text{case1}} \approx (1 - \text{sel}) \cdot (1 - (1 - \text{sel})^\mu).$$

PROOF. By definition, a Case-(1) false positive occurs when the Marker of an edge $e = (u \rightarrow v)$ is activated due to at least one dominated node satisfying the query predicate, while the target node v itself does not satisfy the predicate. Formally, the event is

$$\text{FP}_{\text{case1}}(e) \iff (v.A \not\models R) \wedge (\exists p \in D(e) : p \models R).$$

Conditioned on a fixed dominate size $|D(e)| = k$, independence implies

$$\Pr[v.A \not\models R] = 1 - \text{sel},$$

and

$$\Pr(\exists p \in D(e) : p \models R) = 1 - \Pr(\forall p \in D(e) : p \not\models R) = 1 - (1 - \text{sel})^k.$$

Since the two events are independent, we obtain

$$\Pr(\text{FP}_{\text{case1}} \mid |D(e)| = k) = (1 - \text{sel}) \cdot (1 - (1 - \text{sel})^k).$$

Taking expectation over the random variable $|D(e)|$ yields

$$\Pr(\text{FP}_{\text{case1}}) = (1 - \text{sel}) \cdot \left(1 - \mathbb{E}[(1 - \text{sel})^{|D(e)|}]\right).$$

To derive a bound that depends only on the average dominate size $\mu = \mathbb{E}[|D(e)|]$, observe that the function $f(k) = (1 - \text{sel})^k$ is convex for $k \geq 0$. By Jensen's inequality,

$$\mathbb{E}[(1 - \text{sel})^{|D(e)|}] \geq (1 - \text{sel})^{\mathbb{E}[|D(e)|]} = (1 - \text{sel})^\mu.$$

Substituting this inequality gives

$$\text{FPR}_{\text{case1}} \leq (1 - \text{sel}) \cdot (1 - (1 - \text{sel})^\mu).$$

Non-negativity of probabilities establishes the lower bound. $\square$

Example. At $\text{sel} = 1\%$, the upper bound in Theorem A.6 is approximately 1.97% for $\mu = 2$ and 2.94% for $\mu = 3$. At $\text{sel} = 50\%$, the corresponding bounds are 37.5% and 43.75%. These are upper bounds for a varying dominated-set size; the exact rate equals them when every edge has the stated dominated-set size.

Next, we isolate Codebook discretization for conjunctive predicates on single-valued attributes. This model excludes dominated-node aggregation and set-valued categorical predicates. Let $a_j(v)$ denote the unique value of attribute $A^{(j)}$ in record v .

Let $R = \bigwedge_{j=1}^m R_j$ be a conjunctive predicate on m attributes, and let

$$\mathcal{R} = \{v \in \mathcal{D} \mid R(v) = \text{true}\}$$

denote the set of records satisfying R .

Let $\tilde{\mathcal{R}}_j$ be the union of all Codebook regions containing a value that satisfies R_j . The Codebook-only approximation accepts the records

$$\tilde{\mathcal{R}} = \{v \in \mathcal{D} \mid a_j(v) \in \tilde{\mathcal{R}}_j \text{ for all } j\}.$$

Thus $\mathcal{R} \subseteq \tilde{\mathcal{R}}$. This is not the acceptance set of full, neighborhood-aggregated edge Markers.

Define the Case-(2) false positive rate as

$$\text{FPR}_{\text{case2}}(R) = \frac{\Pr[v \in \tilde{\mathcal{R}} \setminus \mathcal{R}]}{\Pr[v \in \tilde{\mathcal{R}}]}.$$

THEOREM A.7 (BOUNDING CASE-(2) FALSE POSITIVES). Assume that:

- (i) For each attribute $A^{(j)}$, the Codebook partitions its domain into s disjoint regions, each with probability mass at most $1/s$;
- (ii) For each attribute $A^{(j)}$, the predicate R_j partially intersects at most b_j Codebook regions.

If the joint predicate selectivity satisfies $\Pr[v \in \mathcal{R}] \geq \text{sel} > 0$, then

$$\text{FPR}_{\text{case2}}(R) \leq \frac{\sum_{j=1}^m \frac{b_j}{s}}{\text{sel} + \sum_{j=1}^m \frac{b_j}{s}}.$$

THEOREM A.8 (CODEBOOK SIZE FOR CONTROLLING CASE-(2) FALSE POSITIVES). Under the assumptions of Theorem A.7,

$$\text{FPR}_{\text{case2}}(R) \leq \frac{\sum_{j=1}^m \frac{b_j}{s}}{\text{sel} + \sum_{j=1}^m \frac{b_j}{s}}.$$

For a target $0 < \text{FP} < 1$, to guarantee $\text{FPR}_{\text{case2}}(R) \leq \text{FP}$, it suffices that

$$s \geq \frac{(1 - \text{FP})}{\text{FP} \cdot \text{sel}} \cdot \sum_{j=1}^m b_j.$$

PROOF. For a randomly drawn record v , let E_j be the event that $a_j(v) \in \tilde{\mathcal{R}}_j$ but $R_j(a_j(v)) = \text{false}$. Such a value must lie in a Codebook region that intersects the satisfying set of R_j but is not fully contained in it. There are at most b_j such regions, each of probability mass at most $1/s$, so

$$\Pr[E_j] \leq \frac{b_j}{s}.$$

Since

$$\tilde{\mathcal{R}} \setminus \mathcal{R} \subseteq \bigcup_{j=1}^m E_j,$$

the union bound gives

$$\Pr[v \in \tilde{\mathcal{R}} \setminus \mathcal{R}] \leq \sum_{j=1}^m \Pr[E_j] \leq \sum_{j=1}^m \frac{b_j}{s}.$$

This step does not require independence among attributes. Write $t = \Pr[v \in \mathcal{R}] \geq \text{sel} > 0$ and $\delta = \Pr[v \in \tilde{\mathcal{R}} \setminus \mathcal{R}]$. Since $\mathcal{R} \subseteq \tilde{\mathcal{R}}$,

$$\Pr[v \in \tilde{\mathcal{R}}] = t + \delta.$$

The ratio $\delta/(t + \delta)$ increases with δ and decreases with t . Using the preceding bounds gives

$$\text{FPR}_{\text{case2}}(R) = \frac{\delta}{t + \delta} \leq \frac{\sum_{j=1}^m \frac{b_j}{s}}{\text{sel} + \sum_{j=1}^m \frac{b_j}{s}}.$$

Requiring the right-hand side to be at most FP gives

$$(1 - \text{FP}) \sum_{j=1}^m \frac{b_j}{s} \leq \text{FP} \cdot \text{sel},$$

which is equivalent to the stated sufficient condition on s . $\square$

Example. Consider a conjunction $R = \bigwedge_{j=1}^m R_j$ on m single-valued attributes. If each R_j is a 1D range predicate under a contiguous Codebook partition, then $b_j \leq 2$. Then $\sum_{j=1}^m b_j \leq 2m$, and Theorem A.8 gives the sufficient condition

$$s \geq \frac{(1 - \text{FP})}{\text{FP} \cdot \text{sel}} \cdot 2m.$$

For instance, with $m = 4$, $\text{sel} = 50\%$, and $\text{FP} = 50\%$, it suffices to use $s \geq 16$ bits per attribute. If $\text{sel} = 10\%$ and $\text{FP} = 50\%$, the bound becomes $s \geq 80$.

These theorems establish the computational and theoretical foundations of EMA. We analyze the index-time and space complexity of EMA, and provide a principled characterization of false positives arising from dominance-based aggregation (Case 1) and codebook granularity (Case 2). Together, these results clarify when marker-based pruning is effective and show how appropriate codebook sizing can bound granularity-induced false positives while preserving efficient graph traversal.

B LOW SELECTIVITY PERFORMANCE UNDER BIASED SELECTIVITY DISTRIBUTIONS.

In practice, multi-predicate queries do not always exhibit evenly distributed selectivities across predicates. We therefore evaluate performance under biased selectivity distributions for range+range queries. Figure 1 reports the results at 95% recall, where the selectivity of the first predicate varies from 55% to 80%, while the selectivity of the second predicate ranges from 1.82% to 12.5%.

Compared to evenly distributed queries, EMA exhibits a slight performance degradation under biased selectivity distributions, but still achieves the best performance in most cases. This demonstrates that EMA remains robust even when selectivity is highly skewed across predicates. Predicate-agnostic methods such as ACORN and NaviX are not affected by biased selectivity distributions; however, they still achieve relatively low QPS in these settings.

We additionally include iRangeGraph in this evaluation as a strong baseline for multi-predicate range queries. iRangeGraph benefits from a high-quality sub-index on the first attribute; however, when the first predicate becomes relatively loose, its search behavior increasingly resembles post-filtering, which limits its effectiveness compared to EMA. iRangeGraph failed to construct its index within the 20-hour time limit on Wiki and is not included there.

On the YouTube-RGB dataset, Milvus outperforms EMA, primarily because the dataset size is relatively small and Milvus can effectively leverage parallel linear scan under highly restrictive predicates. This advantage reflects Milvus's internal multi-threaded execution.

Table 1: HashANN QPS at recall=0.95 on Redcaps_4M across selectivity and graph degree M .

M	1%	3%	10%	40%	80%	100%
16	375	635	931	2200	2958	3765
32	308	591	1112	2606	3097	4904
40	330	576	1073	2575	3436	5438
64	286	533	1056	2364	3399	4870
best M	16	16	32	32	40	40

C FAIR COMPARISON WITH HNSWLIB AND NAVIX

Since HNSWLib natively supports filtering, we include it as a baseline to demonstrate the effectiveness of EMA. In addition, NaviX is implemented on top of Faiss, which is generally less competitive than HNSWLib. To ensure a fair comparison, we evaluate (i) naive filtering in HNSWLib and (ii) NaviX implemented on top of HNSWLib.

Figure 2a shows that EMA consistently outperforms HNSWLib. Moreover, NaviX does not achieve better performance than the HNSWLib baseline.

Figure 2b reports recall-QPS trade-offs at 10% selectivity. We implement a prototype of NaviX on HNSWLib. Although HNSWLib alone achieves competitive performance, NaviX is sensitive to the underlying index behavior. Compared to Faiss, HNSWLib explores more candidates at the bottom layer; the additional two-hop expansion further amplifies random memory accesses, leading to degraded performance. A detailed analysis will be included in the revision.

D PARAMETER SENSITIVITY ANALYSIS

D.1 Effect of Graph Degree M

We evaluate the impact of the maximum graph degree $M \in \{16, 32, 40, 64\}$. All other parameters are fixed ($\text{ft_bits} = 128$, $\text{ft_routing_min_deg} = 8$, $e_{fc} = 300$), using single-threaded execution over 1,000 queries. Table 1 reports QPS at 95% recall, where bold entries indicate the best M for each selectivity.

Overall, the optimal choice of M depends on selectivity. At low selectivity (1%–3%), smaller M performs best, as it reduces neighbor comparison cost and enables faster transition to edge recovery. Increasing M brings no benefit in this regime due to limited candidate expansion. At moderate selectivity (10%–40%), $M = 32$ achieves the highest QPS, balancing search quality and traversal overhead. At high selectivity (80%–100%), $M = 40$ becomes optimal, as slightly larger neighborhoods improve connectivity without incurring excessive scanning.

Notably, overly large M (e.g., 64) consistently underperforms due to increased traversal and unnecessary neighbor evaluation. These results suggest that moderate values of M provide the best trade-off between exploration cost and graph connectivity across selectivity levels.

D.2 Effect of FT Width (ft_bits)

We evaluate the impact of ft_bits on performance. Figure 3 reports QPS at 95% recall across selectivity.

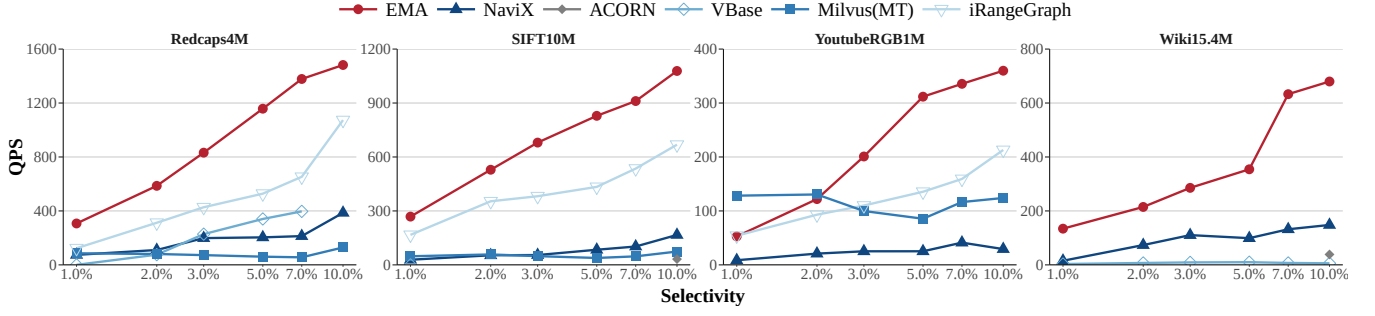


Figure 1: QPS for range+range biased multi-predicate query at 1%-10% selectivity with 95% recall@10.

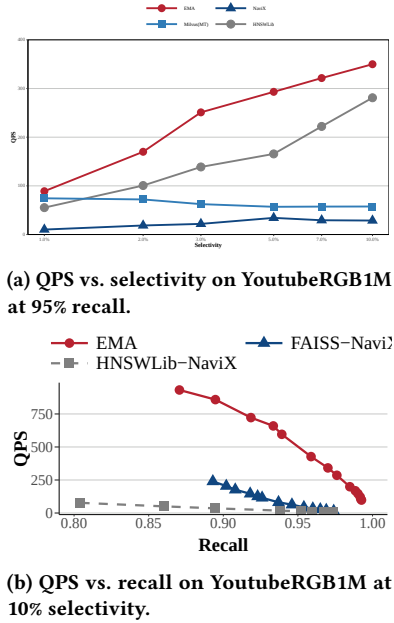


Figure 2: Experiments across different baselines.

$ft_bits=128$ consistently achieves the best performance, improving over 32 bits by 2%–25%. Increasing from 32 to 128 bits reduces false positives and improves routing accuracy, while further increasing to 256 bits yields no benefit due to the enlarged memory footprint, which increases cache access cost and dominates query latency. $ft_bits=64$ provides a reasonable trade-off but remains slightly inferior to 128 bits. Thus, we use $ft_bits=128$ by default.

D.3 Effect of Edge Recovery Threshold (min_deg)

We evaluate the effect of the minimum degree in edge recovery (min_deg). Figure 4 reports QPS at 95% recall under varying selectivity.

Performance exhibits a clear peak at moderate min_deg . Under $M=40$ (level-0 degree ≈ 80), the optimal range lies between 15 and 20, achieving the highest QPS across selectivities. Increasing

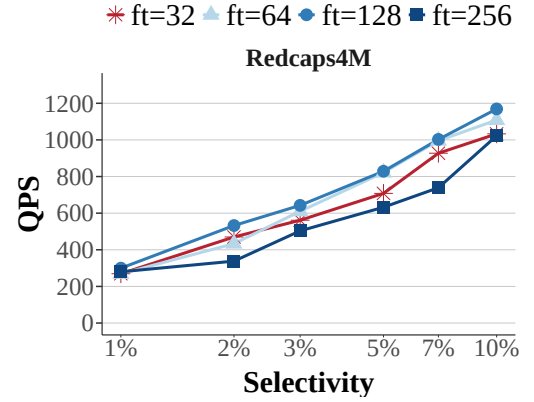


Figure 3: Impact of ft_bits on QPS at 95% recall across different selectivity levels.

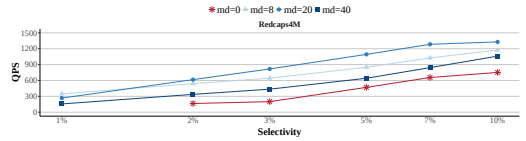


Figure 4: Impact of min_deg on QPS at 95% recall across different min_deg levels.

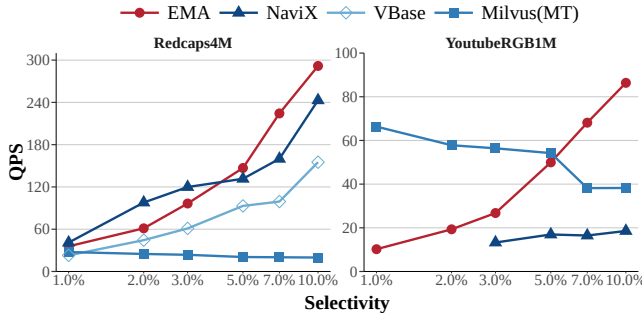
min_deg beyond this range leads to a sharp degradation (30%–55%), as the search progressively approaches post-filtering behavior.

When min_deg becomes large (e.g., ≥ 40), the effective neighborhood is saturated, and traversal degenerates to visiting nearly all neighbors, effectively discarding FT-based routing. This regime is consistently slower (up to 2.9 $\times$) than the optimal setting, highlighting that FT routing is effective only when min_deg is properly constrained.

Overall, these results reveal a narrow operating region ($min_deg \in [8, 20]$) where EMA significantly outperforms post-filtering, confirming that controlled edge recovery is critical to preserving routing efficiency.

Table 2: FPR on Redcaps1M.

Selectivity	FPR	
	Range+Label	Range
1%	3.84%	2.99%
5%	4.80%	3.67%
10%	5.41%	4.27%
20%	6.20%	5.18%
60%	7.97%	7.71%

**Figure 5: Label-range query at 95% recall@100.**

E OTHER STUDIES

E.1 False positive rate of the Marker.

The effectiveness of EMA depends on the accuracy of the Marker, as a lower false positive rate (FPR) reduces unnecessary candidate exploration during joint filtering. Table 2 reports the FPR on RedCaps1M across selectivity levels ranging from 1% to 60%.

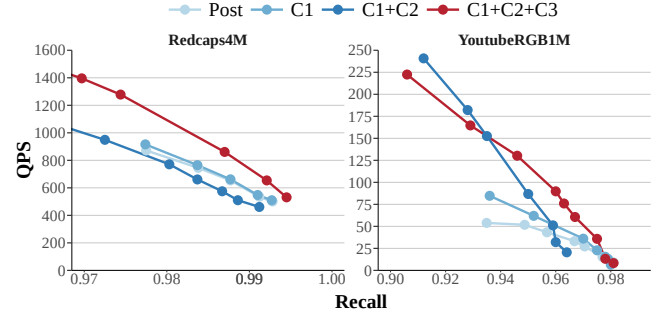
The reported false positive rates are measured with edge recovery disabled and a fixed search parameter (efs), so as to isolate the accuracy of the Marker itself. Across this broad range of selectivities, the FPR consistently remains below 10% for both range+label and range-only queries. As selectivity increases, the FPR grows moderately, which is expected as filtering becomes less restrictive.

Overall, these results indicate that the Marker provides effective pre-checking with a low and stable false positive rate, supporting efficient query processing across different selectivity regimes.

E.2 Performance at 95% Recall@100.

Figure 5 reports the QPS under the $R@100$ setting. While EMA exhibits slightly lower throughput in some cases, it is important to note that EMA consistently achieves a higher recall level (approximately 99%) compared to the 95% recall target used by other methods.

This difference arises from the more conservative convergence criteria adopted by EMA, which prioritize retrieval accuracy and result stability. As a result, the reported throughput reflects a stricter recall requirement, highlighting a trade-off between efficiency and accuracy under the $Rrecall@100$ evaluation.

**Figure 6: Ablation study on Redcaps and Youtube-RGB.**

E.3 Ablation study.

Building upon the classic post-filtering approach on HNSW, we decompose our method into three key components: (1) diversity-aware pruning (C1), (2) joint filtering with Markers (C2), and (3) neighbor recovery under low selectivity (C3).

Figure 6 presents an ablation study on the YouTube-RGB dataset under 1% selectivity and the RedCaps dataset under 10% selectivity with both range and label constraints. Compared with the original HNSW post-filtering baseline, each proposed component consistently improves retrieval performance, and their combination yields the best overall results. In particular, introducing the Marker slightly reduces QPS at high recall targets, but yields a noticeable performance improvement at lower recall levels (around 92%). This improvement indicates that the Marker effectively prunes unnecessary traversals while preserving sufficient candidates, providing a strong foundation for further performance gains when combined with the edge recovery strategy.

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